Notes on Algebra

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1 Gaussian Elimination

For an augmented matrix that represents a one-to-one correspondence with linear equations.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \longleftrightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}, \tag{1}$$

Assuming the base field $\mathbb{F}$ so that nonzero elements are invertible, we define the transformations

Definition 1.1 (I, II: Elementary Row Operations)**.**

$$I(i,k): \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \longmapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{pmatrix}.$$

$$II(i,k,c): \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \longmapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1}+ca_{k1} & a_{i2}+ca_{k2} & \cdots & a_{in}+ca_{kn} & b_i+cb_k \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix}.$$

Conducting operations of type I, II, the resulting augmented matrix represents a linear system that is equivalent to the original one.

Gaussian elimination can be standardized for an augmented matrix A,

$$G(A): \text{if } a_{11} = 0 \text{ and } \exists i > 1, a_{i1} \neq 0,\ I(1,i).$$
$$\forall 1 < i \leqslant n,\ II(i, 1, -a_{i1}a_{11}^{-1}).$$

We iterate $G(A)$, with the results matrix

$$\bar{A} := \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \cdots & \bar{a}_{1n} & b'_1 \\ 0 & \cdots & \bar{a}_{2i} & \bar{a}_{2(i+1)} & \bar{a}_{2n} & b'_2 \\ 0 & \cdots & \bar{a}_{3k} & \bar{a}_{3(k+1)} & \bar{a}_{3n} & b'_3 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \bar{a}_{rs} & \bar{a}_{r(s+1)} & \bar{a}_{rn} & b'_r \\ 0 & 0 & 0 & \cdots & 0 & b'_{r+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & b'_m \end{pmatrix}$$

in which $1 < i < k < \cdots < s$. $\bar{a}_{ij}$ denote the entries after elimination.

Theorem 1.2. *The conditions of linear equations ascertaining* consistency *and* determinacy *are as follows.*

1. *If rows such as $(0, 0, \cdots, b_i \neq 0)$ do not exist in the result matrix of Gaussian elimination, then the linear equations are consistent.*

2. *If $r = n$ for consistent linear equations, they are determined.*

Proof. We mimic the regression process, observing the result matrix. The main unknown number x_{rs} is determined by other free unknown numbers $x_{r(s+1)}, x_{r(s+2)}, \cdots, x_{rn}$. Similarly, $x_{(r-1)k}$ is determined by x_{rs} and $x_{r(s+1)}, x_{r(s+2)}, \cdots, x_{rn}$. Hence, the system is consistent.

If $r = n$, then there are no free unknown numbers, and thus, the linear equations are determined, which means

Consistent linear equations are determined $\Longleftrightarrow$ the result matrix in *Row Echelon Form (REF)* satisfies $r = n$.

□

From the perspective of mapping, we define

$$T_A : V \to W, \quad \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \longmapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x} \to A\boldsymbol{x}}. \tag{2}$$

The result we summarized in Gaussian elimination can be transcribed as:

$$\ker T_A = \ker T_{G(A)} \quad \text{(When } b_i = 0\text{, the } G(A) \text{ process doesn't alter the solutions of the system)} \tag{3}$$
$$\operatorname{Im} T_A = \operatorname{Im} T_{G(A)} \quad \text{(one-to-one correspondence can be established through the definition of } II\text{)} \tag{4}$$

Furthermore, by Gaussian elimination, we can conclude that the matrix for every T_A is row-equivalent to one *Row Echelon Form* $\bar{A}$.

$$\begin{aligned} &\forall T_A : V \to W,\ \boldsymbol{x} \mapsto A\boldsymbol{x}, \\ \Longleftrightarrow\ & T_{G(A)} : V \to W, \boldsymbol{x} \mapsto G(A)\boldsymbol{x}. \end{aligned}$$

If the $r = n$, then the matrix for T_A is row-equivalent to one REF matrix such that there are no $(0, 0, \cdots, b_i)$ rows.

2 Matrices & Determinants

Definition 2.1 (Determinants)**.** *The $n-$order determinants are*

$$\det A = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} := \sum_{i=1}^{n} (-1)^{i+1} a_{i1} M_{i1}.$$

M_{ij}, the minor, is a $n-1$-order determinant that excludes the i^{th} row and j^{th} column of A. From the definition,

1. *$\det A$ is a linear function for every row of A*
2. *$A \xrightarrow{I(i,k)} A', \det A = -\det A'$*

Theorem 2.2.

$$\begin{aligned} A &\xrightarrow{I(i,k)} A',\ \det A = -\det A' \\ A &\xrightarrow{II(i,k)} A',\ \det A = \det A'. \end{aligned}$$

Proof. The proof is followed by the fact that the determinants of matrix A can be expressed as a linear function of any $A's$ row. It is solvable by induction. □

Theorem 2.3. *From the last theorem, we discuss the effect of Gaussian elimination on determinants. For matrix $r = n$,*

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \xLeftrightarrow[I,\, II]{\textit{row-equivalent}} \bar{A} = \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \bar{a}_{1n} \\ 0 & \bar{a}_{22} & \cdots & \bar{a}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \bar{a}_{nn} \end{pmatrix}$$

1. $\det A = \bar{a}_{11}\bar{a}_{22}\bar{a}_{33}\bar{a}_{44}\cdots\bar{a}_{nn}$.
2. *The linear equations with the coefficient matrix A are consistent and determined* $\iff \det A \neq 0$.

Definition 2.4 (Singular matrix)**.** *If a matrix's determinant is* 0, *then it is singular.*

From these attributes of determinants, we are able to deduce the following.

Theorem 2.5. $\forall F : M_{m\times n}(\mathbb{R}) \to \mathbb{R},\ A \mapsto F(A)$ *satisfies:*

1. *$F(A)$ is a linear function for every row of A,*
2. $A \xrightarrow{I(i,k)} A', F(A) = -F(A')$.

$F(A) = F(E)\det A$.

Proof. The proof of the theorem is back substitution. We always reduce A to an upper triangular matrix. We first notice that from $A \xrightarrow{I(i,k)} A', F(A) = -F(A')$, we obtain that $F(A) = (-1)^t F(\bar{A})$ if we use I process t times since II doesn't alter $F(A)$. Thus, what's left is

$$F(\bar{A}) = F(E)\bar{a}_{11}\bar{a}_{22}\bar{a}_{33}\cdots\bar{a}_{nn},$$

in which

$$E := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix},\ \det E = 1.$$

Since $F(A)$ is a linear function for every row of A,

$$\begin{aligned} F(\bar{A}) &= \bar{a}_{nn} \cdot F\left(\begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \bar{a}_{1n} \\ 0 & \bar{a}_{22} & \cdots & \bar{a}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}\right) \\ \xRightarrow{\forall 1\leqslant i<n,\ II(i,n,-\bar{a}_{in})} &= \bar{a}_{nn} \cdot F\left(\begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & 0 \\ 0 & \bar{a}_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}\right). \end{aligned}$$

The process is a replica of back substitution. Matrix $\bar{A}$ would eventually be E. □

The Column of the determinant

We consider the propagated form of the determinant's definition, which consists of eliminating a random column of the original matrix:

$$F_j(A) = \sum_{i=1}^{n} (-1)^{i+1} a_{ij} M_{ij}.$$

We can simply conclude that it satisfies Theorem 2.5; thus $F_j(A) = F_j(E)\det A$.

$$\begin{aligned}
F_j(E) &= \sum_{i=1}^{n}(-1)^{i+1}a_{ij}M_{ij} = (-1)^{j+1}a_{jj}M_{jj} = (-1)^{j+1} \\
F_j(A) &= (-1)^{j+1}\det A \\
\det A &= \sum_{i=1}^{n}(-1)^{i+j}M_{ij}\cdot a_{ij} =: \sum_{i=1}^{n}A_{ij}a_{ij}.
\end{aligned}$$

by induction hypothesis: $M_{jj} = \det E$ for order $n-1$, $\det E = 1$. A_{ij} is the *Algebraic cofactor* of a_{ij}. The linearity of determinants for columns, with the assistance of algebraic cofactors, implies the viability of Theorem 2.2 on columns.

Lemma 2.6 (algebraic cofactors acting on linear combinations)**.**

$$\forall k \neq j,\ \sum_{i=1}^{n}A_{ik}a_{ij} = 0$$

Proof. We construct a matrix A' such that its k^{th} column is identical to the j^{th} column.
Now, $\det A' = \sum_{i=1}^{n}A'_{ik}a_{ik} = 0$, but the $A'_{ik} = A_{ik}, a_{ij} = a_{ik}$. □

Theorem 2.7 (Cramer's Rule)**.** *For linear equations with the augmented matrix*

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{pmatrix},\ \textit{in which } A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

the k^{th} unknown number

$$x_k = \frac{\det A_k}{\det A}.$$

The A_k is constructed by replacing the k^{th} column of A to $(b_1, b_2, \cdots, b_n)$. If $\det A = 0$, it is a singular matrix and the linear equations is undetermined or inconsistent.

Proof. For x_k, we may multiply

$$\begin{pmatrix} A_{1k}a_{11} & A_{1k}a_{12} & \cdots & A_{1k}a_{1n} \\ A_{2k}a_{21} & A_{2k}a_{22} & \cdots & A_{2k}a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{nk}a_{n1} & A_{nk}a_{n2} & \cdots & A_{nk}a_{nn} \end{pmatrix}, \quad \begin{pmatrix} A_{1k}b_1 \\ A_{2k}b_2 \\ \vdots \\ A_{nk}b_n \end{pmatrix}.$$

Then, we add every term of the linear equations. Every $\sum_{i=1}^{n}A_{ik}a_{ij}$ would cancel out. The only remainder term is x_k, and its coefficient is $\sum_{i=1}^{n}A_{ik}a_{ik} = \det A$. The right-hand side is $\sum_{i=1}^{n}A_{ik}b_i = \det A_k$. □

2.1 Permutations

Definition 2.8 (even(odd) permutation)**.** *Defining the function $\varepsilon(\boldsymbol{J})$, $\boldsymbol{J} = (i_1, i_2, \cdots, i_n)$ is a permutation.*

$$\varepsilon(\boldsymbol{J}) := \begin{cases} 1 & \textit{permutation } \boldsymbol{J} \textit{ is derived by even-numbered transposition from origin} \\ -1 & \textit{permutation } \boldsymbol{J} \textit{ is derived by odd-numbered transposition from origin} \end{cases}$$

If $\varepsilon(\boldsymbol{J}) = 1$, $\boldsymbol{J}$ is an even permutation, and vise versa.

Definition 2.9 (symmetric(antisymmetric) functions)**.** *For a symmetric function $F(\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a_n})$,*

$$F(\boldsymbol{a}_1, \cdots, \boldsymbol{a}_i, \cdots, \boldsymbol{a}_j, \cdots, \boldsymbol{a_n}) = F(\boldsymbol{a}_1, \cdots, \boldsymbol{a}_j, \cdots, \boldsymbol{a}_i, \cdots, \boldsymbol{a_n}).$$

For an antisymmetric function $F(\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a_n})$,

$$F(\boldsymbol{a}_1, \cdots, \boldsymbol{a}_i, \cdots, \boldsymbol{a}_j, \cdots, \boldsymbol{a_n}) = -F(\boldsymbol{a}_1, \cdots, \boldsymbol{a}_j, \cdots, \boldsymbol{a}_i, \cdots, \boldsymbol{a_n}).$$

Theorem 2.10 (A definition of determinant)**.** *The determinant of a nth-order matrix is a solitary function $F(\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a_n})$ of n rows with n elements such that*

1. *It is a linear function for each row*
2. *It is antisymmetric*
3. $F(\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e_n}) = 1$, *in which $\boldsymbol{e}_i$ is the row that has 1 in its ith position and 0 for its other positions.*

Theorem 2.11 (multilinear function)**.** *Any multilinear function can be expressed as*

$$F(\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a_n}) = \sum_{(i_1, i_2, \cdots, i_n)} \alpha_{i_1, i_2, \cdots, i_n} a_{1i_1} a_{2i_2} a_{3i_3} \cdots a_{ni_n}$$

in which $\boldsymbol{a}_i = (a_{i1}, a_{i2}, a_{i3}, \cdots, a_{in})$, and coefficient $\alpha_{i_1, i_2, \cdots, i_n}$ are only related to their indexes rather than rows.
The coefficient $\alpha_{i_1, i_2, \cdots, i_n} = F(\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \cdots, \boldsymbol{e}_{i_n})$.

(idea) It is easily proven by induction, since any linear function is $F(\boldsymbol{x}) = \sum_{i=1}^{n} a_i x_i$. In fact, from Theorem 2.11, we acknowledge the explicit formula for determinants. The determinant function

$$F(\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \cdots, \boldsymbol{e}_{i_n})$$

can be considered a permutation $\boldsymbol{J}$ of the rows from the original matrix E. Since we know the I process adds $-$ to its determinant, we have $F(\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \cdots, \boldsymbol{e}_{i_n}) = \varepsilon(\boldsymbol{J}) \det E = \varepsilon(\boldsymbol{J})$. Thus,

$$\det A = \sum_{\boldsymbol{J}} \varepsilon(\boldsymbol{J}) a_{1i_1} a_{2i_2} a_{3i_3} \cdots a_{ni_n}.$$

2.2 Rank

Definition 2.12 (Rank of a Matrix)**.** *Let $A \in M_{n\times n}$. The* rank *of A, denoted by* $\operatorname{rank}(A)$*, is defined as*

$$\operatorname{rank}(A) = r$$

if and only if:

1. *There exists at least one nonzero $r \times r$ minor of A;*
2. *Every $(r+1) \times (r+1)$ minor of A is zero.*

Equivalently, $\operatorname{rank}(A)$ *is the largest integer r for which A has a nonzero r-th order minor.*

Theorem 2.13. *Let $A \in M_{m\times n}(\mathbb{F})$ and let B be obtained from A by the elementary row operation*

$$II(i, j, p).$$

Then

$$\operatorname{rank}(A) = \operatorname{rank}(B).$$

Proof. Write

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \xrightarrow{II(i,j,p)} B = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} + pa_{j1} & a_{i2} + pa_{j2} & \cdots & a_{in} + pa_{jn} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Assume

$$\operatorname{rank}(A) = r.$$

Then there exists a $r \times r$ nonzero minor

$$M = \begin{vmatrix} a_{i_1 j_1} & \cdots & a_{i_1 j_r} \\ \vdots & & \vdots \\ a_{i_r j_1} & \cdots & a_{i_r j_r} \end{vmatrix} \neq 0,$$

where

$$\{i_1, \dots, i_r\} \subset \{1, \dots, m\}, \quad \{j_1, \dots, j_r\} \subset \{1, \dots, n\}.$$

Case 1: The minor does not involve row i.
Then the corresponding minor in B is unchanged, hence still nonzero. Therefore,

$$\operatorname{rank}(B) \geqslant r.$$

Case 2: Every nonzero r-minor of A contains row i.
Choose such a minor M and assume $i = i_k$ for some k.
Consider the corresponding minor in B:

$$N = \begin{vmatrix} b_{i_1 j_1} & \cdots & b_{i_1 j_r} \\ \vdots & & \vdots \\ b_{i_r j_1} & \cdots & b_{i_r j_r} \end{vmatrix}.$$

Since rows

$$\boldsymbol{b}_i = \boldsymbol{a}_i + p\boldsymbol{a}_j,$$

and the determinant is linear in each row, we obtain

$$N = M + pM',$$

where M' is obtained from M by replacing the i-th row with the j-th row:

$$M' = \begin{vmatrix} a_{i_1 j_1} & \cdots & a_{i_1 j_r} \\ \vdots & & \vdots \\ a_{j j_1} & \cdots & a_{j j_r} \\ \vdots & & \vdots \\ a_{i_r j_1} & \cdots & a_{i_r j_r} \end{vmatrix}.$$

Now we analyze M'. If $j \in \{i_1, \dots, i_r\}$, then M' has two identical rows; hence $M' = 0$.
If $j \notin \{i_1, \dots, i_r\}$, then the row index set of M' is

$$\{i_1, \dots, i_{k-1}, j, i_{k+1}, \dots, i_r\},$$

which does not contain i.

By assumption, every nonzero r-minor must contain row i. Therefore, any r-minor not containing row i must vanish. Hence $M' = 0$.

In both cases, $N = M \neq 0$. Thus, $\operatorname{rank}(B) \geqslant r$. Since the row operation is invertible (its inverse is $II(i, j, -p)$), the same argument gives $\operatorname{rank}(A) \geqslant \operatorname{rank}(B)$. Therefore,

$$\operatorname{rank}(A) = \operatorname{rank}(B).$$

$\square$

From now on, we will use index-set minor notation:

$$A[I \mid J] = \begin{pmatrix} a_{i_1 j_1} & a_{i_1 j_2} & \cdots & a_{i_1 j_r} \\ a_{i_2 j_1} & a_{i_2 j_2} & \cdots & a_{i_2 j_r} \\ \vdots & \vdots & & \vdots \\ a_{i_r j_1} & a_{i_r j_2} & \cdots & a_{i_r j_r} \end{pmatrix}, \quad I = \{i_1, \dots, i_r\}, \quad J = \{j_1, \dots, j_r\}.$$

With a little observation, we noticed that the rank(A) is exactly the number of non-zero rows in the resulting REF matrix of A:

$$\bar{A} = \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \cdots & \bar{a}_{1n} & b'_1 \\ 0 & \cdots & \bar{a}_{2i} & \bar{a}_{2(i+1)} & \bar{a}_{2n} & b'_2 \\ 0 & \cdots & \bar{a}_{3k} & \bar{a}_{3(k+1)} & \bar{a}_{3n} & b'_3 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \bar{a}_{rs} & \bar{a}_{r(s+1)} & \bar{a}_{rn} & b'_r \\ 0 & 0 & 0 & \cdots & 0 & b'_{r+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & b'_m \end{pmatrix}, \quad \operatorname{rank}(\bar{A}) = \operatorname{rank}(A) = r.$$

From the equation, we can improve Theorem 1.2 to its ranked version:

Theorem 2.14 (Improve of Theorem 1.2)**.** *Set*

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \textit{Augmented Matrix } (A \mid B) := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix}$$

The conditions of linear equations ascertaining consistency *and* determinacy *are as follows.*

1. *If* rank(A) = rank($A \mid B$)*, then the linear equations are consistent.*
2. *If* rank(A) $= n$ *for consistent linear equations, they are determined.*

The rank(A) of matrix A unfolds the amount of linear information from linear equations with coefficient A. Rank = number of independent rows.

2.3 The operations of Matrix

Definition 2.15 (Transposition)**.**

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

$$\Longrightarrow A^T := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}, \quad (A^T)_{ij} = a_{ji}.$$

Definition 2.16 (Multiplication)**.**

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \ B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1l} \\ b_{21} & b_{22} & \cdots & b_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nl} \end{pmatrix}, \ l \leqslant m.$$

Then,

$$(AB)_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj} \overset{Einstein}{=:} a_{ik} b_{kj}.$$

The multiplication of Matrices resembles Tensor Contraction*, which is also conducted by taking the sum of one identical index of two multiplying tensors*

$$c_{i_1 i_2 \cdots i_{n-1}} = \sum_{i_k} a_{i_1 i_2 \cdots i_k \cdots i_n} b_{i_1 i_2 \cdots i_k \cdots i_n}.$$

Theorem 2.17. *The multiplication of matrices is associative and distributive.*

$$ABC = A(BC), A(B + C) = AB + AC, (A + B)C = AC + BC.$$

$$(AB)^T = B^T A^T.$$

Lemma 2.18 (Elementary row operation in terms of Matrix Multiplication)**.**

$$I(i,j) \Longleftrightarrow T_{ij} := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & \mathbf{0} & \cdots & \mathbf{1} & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & \mathbf{1} & \cdots & \mathbf{0} & \cdots & 0 \\ \vdots & \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}, \quad (T_{ij})_{rs} = \begin{cases} 1 & r = s \neq i, j \\ 1 & (r,s) = (i,j) \ or \ (j,i) \\ 0 & otherwise \end{cases} \tag{5}$$

$$II(i,j,c) \Longleftrightarrow U_{ij}(c) := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 & \cdots & \boldsymbol{c} & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}, \quad (U_{ij}(c))_{rs} = \begin{cases} 1 & r = s \\ c & (r,s) = (i,j) \\ 0 & otherwise \end{cases} \tag{6}$$

Put simply, we normalized the manipulations for rows to merely left multiplication of a combination of $T_{ij}, U_{ij}(c)$.

Corollary 2.19. *By left multiplying a series of $T_{ij}, U_{ij}(c)$, we are able to reduce a random matrix A to its REF. Moreover,*

$$\operatorname{rank}(A) = \operatorname{rank}(T_{ij}A) = \operatorname{rank}(U_{ij}(c)A)\,.$$

Theorem 2.20. $\forall A, B \in M_{n\times n}, \det AB = \det A \det B.$

Proof. Fixed B, we consider $F(A) = \det AB$. The sole issue is to prove its antisymmetry.

$$(AB)_{ij} = \sum a_{ik}b_{kj}. \text{ For row } i,\ \boldsymbol{c}_i = \left(\sum a_{ik}b_{k1}, \sum a_{ik}b_{k2}, \sum a_{ik}b_{k3}, \cdots, \sum a_{ik}b_{kn}\right).$$

$\Longrightarrow AB$ is multilinear for rows of A.

If we switch the i, j rows of A ($I(i,j)$), it is equivalent to switch i, j rows of AB.

$$\boldsymbol{c}_i = \left(\sum a_{jk}b_{k1}, \sum a_{jk}b_{k2}, \sum a_{jk}b_{k3}, \cdots, \sum a_{jk}b_{kn}\right).$$

$\Longrightarrow F(A) = -F(T_{ij}A)$.Antisymmetry is proved.

Thus, we have $F(A) = F(E)\det A$, but $F(E) = \det B$. ☐

Definition 2.21 (Inverse matrix)**.** *For $A, B \in M_{n\times n}$, if $AB = E$, then $B =: A^{-1}$ is the inverse matrix of A.*

From the definition, we immediately obtain

1. $\det A \det B = \det E = 1$.
2. $\exists! A^{-1}$.
3. $\exists C \in M_{n\times n}, CA = E$. *In fact,* $C = A^{-1}$.

For the third property of the inverse matrix, we present $A^T(A^T)^{-1} = E$. Considering the transposition of the equation $((A^T)^{-1})^T A = E$. $C = ((A^T)^{-1})^T$ exists. Since the association law is valid, uniqueness is obvious.

Theorem 2.22 (Explicit formula of Inverse Matrix)**.** *We use Cramer's Rule. Considering the vector multiplication of the inverse matrix $AB = E$,*

$$A\boldsymbol{b}_j = \boldsymbol{e}_j := (0, 0, \cdots e_j = 1, \cdots, 0)^T.$$

This is a set of linear equations with the same quantity of unknown numbers, so the elements within the inverse matrix is clear:

$$b_{ij} = \frac{\det A_i}{\det A} = \frac{A_{ji}}{\det A}.$$

The A_i is constructed by replacing the i^{th} column of A with $\boldsymbol{e}_j$.

We summarize the Gauss elimination: a $n \times n$ non-singular matrix A can be reduced to REF by left multiplying $U_{ij}(c), T_{ij}$. In addition, we notice that

$$\begin{aligned} &\begin{pmatrix} \boldsymbol{c}_i \\ \boldsymbol{c}_j \end{pmatrix} \xrightarrow{U_{ij}(1)} \begin{pmatrix} \boldsymbol{c}_i + \boldsymbol{c}_j \\ \boldsymbol{c}_j \end{pmatrix} \xrightarrow{U_{ji}(-1)} \begin{pmatrix} \boldsymbol{c}_i + \boldsymbol{c}_j \\ -\boldsymbol{c}_i \end{pmatrix} \xrightarrow{U_{ij}(1)} \begin{pmatrix} \boldsymbol{c}_j \\ -\boldsymbol{c}_i \end{pmatrix} \xrightarrow{V_j(-1)} \begin{pmatrix} \boldsymbol{c}_j \\ \boldsymbol{c}_i \end{pmatrix} \\ \Longrightarrow\ & T_{ij} = V_j(-1)U_{ij}(1)U_{ji}(-1)U_{ij}(1), \end{aligned}$$

where

$$III(i, \alpha) \Longleftrightarrow V_i(\alpha) := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & \alpha & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 & 1 \end{pmatrix}, \quad (V)_{ii} = \alpha.$$

It multiplies α to the ith row of the matrix A.

Reducing the square matrix to REF, which is upper triangular form, we left multiply

$$\prod_{j=1}^{n} \prod_{i \leqslant j} U_{ij}(-\bar{a}_{ij}\bar{a}_{jj}^{-1}).$$

The product eliminates elements that are not on the diagonal. However, the diagonal matrix

$$\bar{A} = \begin{pmatrix} \bar{a}_{11} & 0 & \cdots & 0 \\ 0 & \bar{a}_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \bar{a}_{nn} \end{pmatrix} = \prod_{i=1}^{n} V_i(\bar{a}_{ii})E.$$

Thus, we have concluded that:

Theorem 2.23. $\forall A \in M_{n\times n}, \det A \neq 0$, A^{-1} *is a series of products of $V_i(\alpha), U_{ij}(c)$, which is the same as Gaussian elimination.*

Corollary 2.24. *Since $U_{ij}(c), V_i(\alpha)$ are I, II, III elementary row operations (They don't alter the rank of the original matrix), $\forall A \in M_{n\times n}, B \in M_{m\times m}; \det B, \det A \neq 0$, $\forall C \in M_{n\times m}$,*

$$\operatorname{rank}(C) = \operatorname{rank}(ACB)\,.$$

3 Vector Space

Definition 3.1 (Vector Space)**.** *The vector space L is a set that satisfies the following properties:*

1. *For $+$, a binary operation, the $(L, +)$ is an* Abel group.
2. *$\forall$scaler α, $\alpha(\boldsymbol{x} + \boldsymbol{y}) = \alpha\boldsymbol{x} + \alpha\boldsymbol{y}$.*
3. *$\forall$scaler α, β, $(\alpha + \beta)\boldsymbol{x} = \alpha\boldsymbol{x} + \beta\boldsymbol{x}$.*

4. $\alpha\beta\boldsymbol{x} = \alpha(\beta\boldsymbol{x})$

5. $\forall \boldsymbol{x} \in L, 1\boldsymbol{x} = \boldsymbol{x}$

We therefore present some supplements to the vector space.

Definition 3.2 (Definitions Involving Subspace)**.**

1.$\forall L' \subset L, s.t. \forall \boldsymbol{x}, \boldsymbol{y} \in L', \alpha\boldsymbol{x} + \beta\boldsymbol{y} \in L'. L'$ *is the* subspace *of* L.
2.L *is the* sum *of its subspaces* $L_1, L_2, \cdots, L_k \iff \forall \boldsymbol{x} \in L, \exists \boldsymbol{x}_1 \in L_1, \boldsymbol{x}_2 \in L_2, \cdots, \boldsymbol{x}_k \in L_k$,
$\boldsymbol{x} = \boldsymbol{x}_1 + \boldsymbol{x}_2 + \cdots + \boldsymbol{x}_k$.
$L = L_1 + L_2 + \cdots + L_k$.
3.L *is the* direct sum *of its subspaces* $L_1, L_2, \cdots, L_k$
$\iff$ *L is the sum of subspaces* $L_1, L_2, \cdots, L_k$. *$\boldsymbol{x}'s$ representation in* $\boldsymbol{x}_1 \in L_1, \boldsymbol{x}_2 \in L_2, \cdots, \boldsymbol{x}_k \in L_k$ *is unique.*
$L = L_1 \oplus L_2 \oplus \cdots \oplus L_k$.

Definition 3.3 (Linear Independence/dependence)**.** *For vectors* $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \cdots, \boldsymbol{x}_n$, *if* $\exists \alpha_1, \alpha_2, \cdots, \alpha_n$ *such that*

$$\alpha_1\boldsymbol{x}_1 + \alpha_2\boldsymbol{x}_2 + \alpha_3\boldsymbol{x}_3 + \cdots + \alpha_n\boldsymbol{x}_n = 0$$

for at least one $\alpha_i \neq 0$. $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \cdots, \boldsymbol{x}_n$, *are* linear dependent.
On the other hand, if

$$\alpha_1\boldsymbol{x}_1 + \alpha_2\boldsymbol{x}_2 + \alpha_3\boldsymbol{x}_3 + \cdots + \alpha_n\boldsymbol{x}_n = 0 \iff \alpha_1, \alpha_2, \cdots, \alpha_n = 0,$$

$\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \cdots, \boldsymbol{x}_n$ *are* linear independent.

Definition 3.4 (Dimension of vector space)**.** *Similar to the* $\operatorname{rank}(A)$ *in matrices, the* dim *is the maximum quantity of linearly independent vectors in vector space L. If* $d = \dim L$, *then* $\forall \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \cdots, \boldsymbol{x}_{d+1}, \exists \alpha_1, \alpha_2, \cdots, \alpha_{d+1}$ *such that*

$$\alpha_1\boldsymbol{x}_1 + \alpha_2\boldsymbol{x}_2 + \alpha_3\boldsymbol{x}_3 + \cdots + \alpha_{d+1}\boldsymbol{x}_{d+1} = 0.$$
$$\alpha_1\boldsymbol{x}_1 + \alpha_2\boldsymbol{x}_2 + \cdots + \alpha_{i-1}\boldsymbol{x}_{i-1} + \alpha_{i+1}\boldsymbol{x}_{i+1} + \cdots + \alpha_{d+1}\boldsymbol{x}_{d+1} = \boldsymbol{x}_i$$

for at least one $\alpha_i \neq 0$. *There is one x_i that is composed by other vectors.*

Lemma 3.5. *As in corollary 2.24,*

$$\dim(\{A\boldsymbol{x} : \boldsymbol{x} \in V\}) = \dim(\{U_{ij}(c)AU^T_{i'j'}(c')\boldsymbol{x} : \boldsymbol{x} \in V\}).$$

3.1 Vector basis and Vector Space Operations

Definition 3.6 (basis)**.** $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$, *such that they are linear independent, and* $\forall \boldsymbol{x} \in L, \boldsymbol{x} = a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + \cdots + a_n\boldsymbol{e}_n$.

We are able to immediately derive the following fact that

Theorem 3.7. *For any group of linear independent vectors* $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_m$, *we can find other* $\boldsymbol{e}_m, \boldsymbol{e}_{m+1}, \cdots, \boldsymbol{e}_n$ *such that they form a basis of L.*

The result is proved by simply adding linear independent vectors repeatedly, and it can be written in terms of vector spaces $\forall L' \subset L, \exists L'' \subset L, L = L' \oplus L''$. The next lemma indicates the dimension's restriction on the vector space.

Lemma 3.8. *Any linear combinations* $\boldsymbol{y}_1, \boldsymbol{y}_2, \cdots, \boldsymbol{y}_m$ *of vectors* $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$ *such that* $m > n$ *are linear dependent.*

Proof. Since the $\boldsymbol{y}_i$ is a linear combination of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$, we have

$$\begin{cases} a_{11}\boldsymbol{x}_1 + a_{12}\boldsymbol{x}_2 + \cdots + a_{1n}\boldsymbol{x}_n = \boldsymbol{y}_1 \\ a_{21}\boldsymbol{x}_1 + a_{22}\boldsymbol{x}_2 + \cdots + a_{2n}\boldsymbol{x}_n = \boldsymbol{y}_2 \\ \cdots \\ a_{m1}\boldsymbol{x}_1 + a_{m2}\boldsymbol{x}_2 + \cdots + a_{mn}\boldsymbol{x}_n = \boldsymbol{y}_m \end{cases}.$$

We have to construct m coefficients $\alpha_1, \alpha_2, \cdots, \alpha_m$ that are not all zero and

$$\alpha_1\boldsymbol{y}_1 + \alpha_2\boldsymbol{y}_2 + \cdots + \alpha_m\boldsymbol{y}_m = 0. \tag{7}$$

If we plug (8) into the linear combinations, we observe

$$(a_{11}\alpha_1 + a_{21}\alpha_2 + \cdots + a_{m1}\alpha_m)\boldsymbol{x}_1 + (a_{12}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{m2}\alpha_m)\boldsymbol{x}_2 + \cdots + (a_{1n}\alpha_2 + a_{2n}\alpha_2 + \cdots + a_{mn}\alpha_m)\boldsymbol{x}_n = 0.$$

The result yields that αs ensure the validity of the equation (it equals 0), thus we must have

$$\begin{cases} a_{11}\alpha_1 + a_{21}\alpha_2 + \cdots + a_{m1}\alpha_m = 0 \\ a_{12}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{m2}\alpha_m = 0 \\ \cdots \\ a_{1n}\alpha_2 + a_{2n}\alpha_2 + \cdots + a_{mn}\alpha_m = 0 \end{cases}.$$

We solve αs by Theorem 1.2 since there are fewer equations (m) than unknown numbers (n). □

Corollary 3.9. *If $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$ are bases, then* $\dim L = n$.

Next, we follow the train of thought that the bases are correlated with dimensions.

Theorem 3.10. *L_1, L_2 is a vector space.*

$$\dim(L_1 \oplus L_2) = \dim L_1 + \dim L_2,$$

$$\dim(L_1 + L_2) = \dim L_1 + \dim L_2 - \dim(L_1 \cap L_2).$$

Proof. Let $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n$ and $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_m$ denote a basis of L_1, L_2. Then, every vector in $L_1 \oplus L_2$ is

$$\boldsymbol{x}_1 + \boldsymbol{x}_2 = a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + \cdots + a_n\boldsymbol{e}_n + a_{n+1}\boldsymbol{f}_1 + a_{n+2}\boldsymbol{f}_2 + \cdots + a_{n+m}\boldsymbol{f}_m,$$

$\boldsymbol{x}_1 \in L_1, \boldsymbol{x}_2 \in L_2$. $\boldsymbol{x}_1 + \boldsymbol{x}_2 = 0$ yields the $a_1, a_2, \cdots, a_{n+m} = 0$ since $\boldsymbol{x}_1, \boldsymbol{x}_2 = 0$, which indicates the linear independence of vectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n$ and $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_m$.

The second result can be derived if we set $L_1 \cap L_2 = L_3, L_1 = L_1' \oplus L_3, L_2 = L_2' \oplus L_3$. □

Definition 3.11. *Let V be a finite-dimensional vector space over a field $\mathbb{F}$.*

A flag *in V is a sequence of subspaces*

$$\mathbf{0} = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_k \subset V$$

such that each inclusion is strict.

If, in addition, $\dim V_i = i$ *for all $i = 0, 1, \ldots, n$, where $n = \dim V$ and*

$$\mathbf{0} = V_0 \subset V_1 \subset \cdots \subset V_n = V,$$

then the flag is called a complete flag.

The series of linear spaces is an additive process of new linear independent vectors.

3.2 Operators

Definition 3.12 (Operator)**.** *The linear transformation, or* operator*, is a map $A : L \to M$ such that*

$$A(\boldsymbol{x}+\boldsymbol{y}) = A(\boldsymbol{x}) + A(\boldsymbol{y}), \quad A(\alpha\boldsymbol{x}) = \alpha A(\boldsymbol{x}).$$

The operator is an embodiment of linear algebra in which it not only fully inherits the attributes of matrix algebra, but also surpasses the axis-dependence and is dedicated to the discussion of the algebraic properties of linear spaces.

The operator corresponds to a transformation by a matrix, while a basis from L, M is selected, respectively. Let $\mathcal{E} = (\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$ and $\mathcal{F} = (\boldsymbol{f}_1, \cdots, \boldsymbol{f}_m)$ be bases of L, M, respectively. For any vector in L

$$\begin{gathered}\boldsymbol{x} = \alpha_1\boldsymbol{e}_1 + \alpha_2\boldsymbol{e}_2 + \cdots + \alpha_n\boldsymbol{e}_n,\\ A(\boldsymbol{x}) = \alpha_1 A(\boldsymbol{e}_1) + \alpha_2 A(\boldsymbol{e}_2) + \cdots + \alpha_n A(\boldsymbol{e}_n).\end{gathered}$$

If the operator transforms the basis of L into

$$\begin{cases} A(\boldsymbol{e}_1) = a_{11}\boldsymbol{f}_1 + a_{21}\boldsymbol{f}_2 + \cdots + a_{m1}\boldsymbol{f}_m \\ A(\boldsymbol{e}_2) = a_{12}\boldsymbol{f}_1 + a_{22}\boldsymbol{f}_2 + \cdots + a_{m2}\boldsymbol{f}_m \\ \cdots \\ A(\boldsymbol{e}_n) = a_{1n}\boldsymbol{f}_1 + a_{2n}\boldsymbol{f}_2 + \cdots + a_{mn}\boldsymbol{f}_m \end{cases}, \qquad [A]_{\mathcal{F}\leftarrow\mathcal{E}} := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

then we have if $A(\boldsymbol{x}) = \beta_1\boldsymbol{f}_1 + \beta_2\boldsymbol{f}_2 + \cdots + \beta_m\boldsymbol{f}_m$,

$$\begin{cases} \beta_1 = a_{11}\alpha_1 + a_{12}\alpha_2 + \cdots + a_{1n}\alpha_n \\ \beta_2 = a_{21}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{2n}\alpha_n \\ \cdots \\ \beta_m = a_{m1}\alpha_1 + a_{m2}\alpha_2 + \cdots + a_{mn}\alpha_n \end{cases} \Longrightarrow [A(\boldsymbol{x})]_{\mathcal{F}} = [A]_{\mathcal{F}\leftarrow\mathcal{E}}[\boldsymbol{x}]_{\mathcal{E}} \tag{8}$$

$[A]_{\mathcal{F}\leftarrow\mathcal{E}}$ is solely determined by the basis $\mathcal{E}, \mathcal{F}$, and the operator itself. By converting to matrix $[A]_{\mathcal{F}\leftarrow\mathcal{E}}$ we obtain the trivial but important result. Here, we have just introduced a crucial notation $[\boldsymbol{x}]_{\mathcal{E}}$; it represents the "coordinate" of an unaltered vector $\boldsymbol{x}$ under a specific basis $\mathcal{E}$. For example, the scalars $\beta_1, \beta_2 \cdots, \beta_m$ in (9) is the coordinates of $A(\boldsymbol{x})$ under $\mathcal{F}$.

Lemma 3.13. *$A : L \to N, B : N \to M$. $(BA)(\boldsymbol{x}) := B(A(\boldsymbol{x}))$. Let $\mathcal{E} = (\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$, $\mathcal{G} = (\boldsymbol{g}_1, \cdots, \boldsymbol{g}_l)$ and $\mathcal{F} = (\boldsymbol{f}_1, \cdots, \boldsymbol{f}_m)$ be bases of L, N, M, respectively. We reached that*

$$(BA)(\boldsymbol{e}_i) = \sum_{j=1}^{m} c_{ij}\boldsymbol{f}_j, \quad c_{ij} = ([B]_{\mathcal{F}\leftarrow\mathcal{G}}[A]_{\mathcal{G}\leftarrow\mathcal{E}})_{ij}.$$

In addition, the $A(B+C) = AB + AC, (A+B)C = AC + BC$.

From the fundamental discussion of operators, we instantly resolve the problem involving any shift in coordinates(basis; they are actually equivalent).

Theorem 3.14 (General Shifts in basis)**.** *Let $\mathcal{E} = (\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$, and $\mathcal{E}' = (\boldsymbol{e}'_1, \cdots, \boldsymbol{e}'_n)$ be two bases of L. Since the shift in bases does not change the value of a vector, we regard the shift in bases as an* Identity Operator *$I_L : L \to L$. Then, the matrices*

$$[I_L]^{-1}_{\mathcal{E}'\leftarrow\mathcal{E}} = [I_L]_{\mathcal{E}\leftarrow\mathcal{E}'}.$$

Proof. We recall the expression in (9). Let $\boldsymbol{x} \in L$. If tuple $\mathcal{E} = (\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n)$ transforms to $\mathcal{E}' = (\boldsymbol{e}'_1, \boldsymbol{e}'_2, \cdots, \boldsymbol{e}'_n)$, then the coordinates alter as

$$[\boldsymbol{x}]_{\mathcal{E}'} = [I_L(\boldsymbol{x})]_{\mathcal{E}'} = [I_L]_{\mathcal{E}'\leftarrow\mathcal{E}}[\boldsymbol{x}]_{\mathcal{E}},$$

which implies that $[I_L]^{-1}_{\mathcal{E}'\leftarrow\mathcal{E}} = [I_L]_{\mathcal{E}\leftarrow\mathcal{E}'}$. ∎

Theorem 3.15 (Operators' matrix with basis Shifts)**.** *$A : L \to M$ is an operator. Let $\mathcal{E} = (\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$, and $\mathcal{E}' = (\boldsymbol{e}'_1, \cdots, \boldsymbol{e}'_n)$ be two bases of L; $\mathcal{F} = (\boldsymbol{f}_1, \cdots, \boldsymbol{f}_n)$, and $\mathcal{F}' = (\boldsymbol{f}'_1, \cdots, \boldsymbol{f}'_n)$ be two bases of M.*

$$\begin{tikzcd}
{[x]_{\mathcal{E}'}} \arrow[d, "A:L\to M"] & {[x]_{\mathcal{E}}} \arrow[l, "I_L:L\to L"'] \arrow[d, "A:L\to M"] \\
{[A(x)]_{\mathcal{F}'}} & {[A(x)]_{\mathcal{F}}} \arrow[l, "I_M:M\to M"']
\end{tikzcd}$$

From the diagram, we acquire the relation of matrices

$$[A]_{\mathcal{F}'\leftarrow\mathcal{E}'} = [I_M]_{\mathcal{F}'\leftarrow\mathcal{F}}[A]_{\mathcal{F}\leftarrow\mathcal{E}}[I_L]^{-1}_{\mathcal{E}'\leftarrow\mathcal{E}}.$$

Proof. The main idea is shown in the commutative diagram. In fact, by Lemma 3.13 and Theorem 3.14, we obtain

$$\begin{cases} [A(x)]_{\mathcal{F}'} = [A]_{\mathcal{F}'\leftarrow\mathcal{E}'}[x]_{\mathcal{E}'} = [A]_{\mathcal{F}'\leftarrow\mathcal{E}'}[I_L]_{\mathcal{E}'\leftarrow\mathcal{E}}[x]_{\mathcal{E}} \\ [A(x)]_{\mathcal{F}} = [A]_{\mathcal{F}\leftarrow\mathcal{E}}[x]_{\mathcal{E}} \\ [A(x)]_{\mathcal{F}'} = [I_M]_{\mathcal{F}'\leftarrow\mathcal{F}}[A(x)]_{\mathcal{F}} \end{cases} \Longrightarrow [A]_{\mathcal{F}'\leftarrow\mathcal{E}'}[I_L]_{\mathcal{E}'\leftarrow\mathcal{E}}[x]_{\mathcal{E}} = [I_M]_{\mathcal{F}'\leftarrow\mathcal{F}}[A]_{\mathcal{F}\leftarrow\mathcal{E}}[x]_{\mathcal{E}}$$

$$\Longrightarrow [A]_{\mathcal{F}'\leftarrow\mathcal{E}'}[I_L]_{\mathcal{E}'\leftarrow\mathcal{E}} = [I_M]_{\mathcal{F}'\leftarrow\mathcal{F}}[A]_{\mathcal{F}\leftarrow\mathcal{E}}$$

Because I is mapping $L \to L$, the matrix $[I_L]_{\mathcal{E}'\leftarrow\mathcal{E}}$ must be a square matrix with a non-zero determinant. Thus, any $[I_L]_{\mathcal{E}'\leftarrow\mathcal{E}}$ is nonsingular. ☐

Corollary 3.16. *We note that if A maps $L \to L$, then the result is reduced to*

$$[A]_{\mathcal{E}'\leftarrow\mathcal{E}'} = [I_L]_{\mathcal{E}'\leftarrow\mathcal{E}}[A]_{\mathcal{E}\leftarrow\mathcal{E}}[I_L]^{-1}_{\mathcal{E}'\leftarrow\mathcal{E}}.$$

The result means that for matrices M', M, P such that $M' = PMP^{-1}$, M, M' belong to one operator.

Remark 3.17. *The book usually employs the other perspective of bases shifts, which is, without introducing I_L, constructing a "shift" operator $P : L \to L$ that directly turns the bases. $P(\boldsymbol{e}_i) = \boldsymbol{e}'_i$. Then, $[P(\boldsymbol{x})]_{\mathcal{E}'} = [\boldsymbol{x}]_{\mathcal{E}}$.*

Definition 3.18 (Determinant of an Operator)**.** *For any operator $A : L \to L$, its determinant is defined as* $\det A := \det[A]_{\mathcal{E}\leftarrow\mathcal{E}}$ *for any basis.*

3.3 Linear Space Algebra

Definition 3.19 (Isomorphism)**.** *If operator $A : L \to M$ is also a bijection (One-to-one correspondence), then it is an isomorphism. We denote $A : L \xrightarrow{\sim} M$, and $L \cong M$. The Isomorphism is an equivalence relation.*

Theorem 3.20. *If linear spaces L, M have limited dimensions, $L \cong M \iff$* $\dim L = \dim M$.

Proof. The necessary condition of the theorem, $\dim L = \dim M \Rightarrow L \cong M$, can be proved by using the $\mathbb{K}^n := \{(a_1, a_2, \cdots, a_n) : a_1, a_2, \cdots, a_n \in \mathbb{K}\}$. It clear that $L \cong \mathbb{K}^n, M \cong \mathbb{K}^n$, and $L \cong M$.

The sufficient condition requires that for a basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$, $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n = A(\boldsymbol{e}_n) \in M$ is also a basis when $L \cong M$, which in turn indicates that $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n$ are linear independent, and $\forall \boldsymbol{x} \in M$, $\boldsymbol{x}$ is a linear combination of $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n$. Let $A \xrightarrow{\sim} M$.

If $a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + \cdots + a_n\boldsymbol{e}_n = \boldsymbol{0}$, then $A(a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + \cdots + a_n\boldsymbol{e}_n) = A(\boldsymbol{0})$, $a_1\boldsymbol{f}_1 + a_2\boldsymbol{f}_2 + \cdots + a_n\boldsymbol{f}_n = \boldsymbol{0}' \in M$, but such a scenario only takes place while $a_1, a_2, \cdots, a_n = 0$ since $A(\boldsymbol{0}) = \boldsymbol{0}'$. Thus, $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n$ are linear independent.

For any $\boldsymbol{x} \in M$, $\exists! \boldsymbol{y} \in L, A(\boldsymbol{y}) = \boldsymbol{x}$. $\boldsymbol{y} = a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + \cdots + a_n\boldsymbol{e}_n$, so $\boldsymbol{x} = A(a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + \cdots + a_n\boldsymbol{e}_n) = a_1\boldsymbol{f}_1 + a_2\boldsymbol{f}_2 + \cdots + a_n\boldsymbol{f}_n$. Thus, $\forall \boldsymbol{x} \in M$, $\boldsymbol{x}$ is a linear combination of $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n$. ☐

Definition 3.21 (Kernel)**.** *For $A : L \to M$,* $\ker A := \{\boldsymbol{x} \in L : A(\boldsymbol{x}) = \boldsymbol{0}' \in M\}$. $\ker A \subset L$.

Definition 3.22 (Image)**.** *For $A : L \to M$,* $\operatorname{Im} A := \{A(\boldsymbol{x}) : \boldsymbol{x} \in L\}$. $\operatorname{Im} A \subset M$.

It is time to introduce the Rank-Nullity theorem to reveal the correlation between our previous insights, which we have already seen in the Gaussian elimination and Rank chapters.

Theorem 3.23 (Rank-Nullity Theorem)**.** *For any linear transformation such that $T : V \to W, \dim V$ is finite,*

$$\dim(\operatorname{Im} T) + \dim(\ker T) = \dim V \tag{9}$$

The creation of the theorem is inspired by (3), (4). We can easily verify some interesting facts: for $[T(\boldsymbol{x})]_{\mathcal{E}'} = A[\boldsymbol{x}]_{\mathcal{E}}, \operatorname{rank}(A) = \dim(\operatorname{Im} T)$; $\dim(\ker T)$ is defined as the *nullity* of the mapping T.

We are able to provide a draft addressing the validity of the Rank-Nullity Theorem *in my words*.

Proof. The linear map T is depicted by a matrix A that transforms the vectors in V to W,

$T : V \to W,\ [\boldsymbol{x}]_{\mathcal{E}} \longmapsto [T]_{\mathcal{E}' \leftarrow \mathcal{E}}[\boldsymbol{x}]_{\mathcal{E}},$

$A := [T]_{\mathcal{E}' \leftarrow \mathcal{E}},\ \boldsymbol{b} := A[\boldsymbol{x}]_{\mathcal{E}}.$

If we choose another basis $\mathcal{F}', \mathcal{E}'$, then according to Theorem 3.15:

$A' = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} A [I_L]^{-1}_{\mathcal{E}' \leftarrow \mathcal{E}} \Longrightarrow \operatorname{rank}(A') = \operatorname{rank}(A)$, which implies the inalterableness of rank.

$\boldsymbol{x} := (x_1, x_2, \cdots, x_n)^T \in V,\ \boldsymbol{b} := (b_1, b_2, \cdots, b_m)^T \in \operatorname{Im} T \subset W,\ A \in M^{m \times n}.$

We shall provide some insight for the proof of the theorem one step at a time.

1. $\dim V = n$, n represents the dimension of the domain V, which is the number of coordinates of the input vector $\boldsymbol{x}$.
2. $\dim(\operatorname{Im} T) = \operatorname{rank}(A)$.

   The $\dim(\operatorname{Im} T)$ represents the number of vectors in a basis in $\operatorname{Im} T = \{A\boldsymbol{x} : \boldsymbol{x} \in \boldsymbol{V}\}$. We incorporate lemma 1.30 to reduce the matrix A to Row Echelon Form, and we notice that the result vector of $\boldsymbol{b}$ is solely determined by the $\operatorname{rank}(A)$ main unknown numbers. Thus, $\dim(\operatorname{Im} T) = \operatorname{rank}(A)$.
3. $\dim(\ker T) =$ the quantity of free unknown numbers in the linear equations $= n - \operatorname{rank}(A)$.

   The $\ker T \subset V$ is a linear subspace. Observing the Row Echelon Form of A, we realize that for each $\boldsymbol{x}$ such that $T(\boldsymbol{x}) = \boldsymbol{0}$, its pivot unknown numbers (the first $\operatorname{rank}(A)$ unknown numbers) are solely determined by its last $n - \operatorname{rank}(A)$ free unknown numbers according to the linear equations. Thus, its dimension is exactly $n - \operatorname{rank}(A)$

Therefore, we obtain $\dim(\operatorname{Im} T) + \dim(\ker T) = \operatorname{rank}(A) + n - \operatorname{rank}(A) = n = \dim V$. □

Remark 3.24. *The rank-nullity theorem is actually a linear-algebraic version of* Isomorphism Theorem A *that puts dimension on both sides.*

$$G/\ker\sigma \cong \operatorname{Im}\sigma \Longrightarrow V/\ker T \cong \operatorname{Im} T, \tag{10}$$

for which

$$\dim V/\ker T = \dim V - \dim(\ker T).$$

My proof is based on rather specific calculations on Matrix; however, the theorem can also be proven in pure operator algebra, refining with extra universality. For such proof to be done, we have to develop our knowledge toward isomorphism.

Theorem 3.25 (Isomorphic Condition involving Kernel)**.**

$$\forall A : L \to M,\ \dim L = \dim M,\ \ker A = \{\boldsymbol{0}\} \Longrightarrow A : L \xrightarrow{\sim} M.$$

Just consider the basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$, and $A(\boldsymbol{e}_1), A(\boldsymbol{e}_2), \cdots, A(\boldsymbol{e}_n) \in M$.

Theorem 3.26 (Isomorphic Condition involving Image)**.**

$$\forall A : L \to M,\ \dim L = \dim M,\ \operatorname{Im} A = M \Longrightarrow A : L \xrightarrow{\sim} M.$$

Consider basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$ such that $A(\boldsymbol{e}_1), A(\boldsymbol{e}_2), \cdots, A(\boldsymbol{e}_n) \in M$ forms a basis.

Theorem 3.27 (Isomorphic Condition involving Singularity)**.** *If for every basis $\mathcal{E} \in L,\ \mathcal{F} \in M,\ A : L \to M$ has* $\det([A]_{\mathcal{F} \leftarrow \mathcal{E}}) \neq 0$, *then A is* nonsingular.

$$A : L \xrightarrow{\sim} M,\ \dim L = \dim M \iff A \textit{ is nonsingular.}$$

Definition 3.28 (Rank of an operator)**.** *For* $A : L \to M$, $\operatorname{rank}(A) := \dim(\operatorname{Im} A)$.

Now, we have enough prerequisites to provide a general proof of the Rank-Nullity theorem.

Theorem 3.29 (Extended Rank-Nullity Theorem)**.** *For an operator* $A : L \to M$, $\operatorname{rank}(A) + \dim(\ker A) = \dim L$. *Plus, we are able to select bases* $\mathcal{E}, \mathcal{F}$ *in* L, M *such that*

$$[A]_{\mathcal{F}\leftarrow\mathcal{E}} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix},\ r := \operatorname{rank}(A)\,.$$

Proof. We decompose $L = \ker A \oplus L_1$, $M = \operatorname{Im} A \oplus M_1$. Considering the restriction of A on $L_1 \to \operatorname{Im} A$, namely $A' : L_1 \to \operatorname{Im} A$, $\forall \boldsymbol{x} \in L, \boldsymbol{x} = \boldsymbol{u} + \boldsymbol{v},\ \boldsymbol{u} \in \ker A, \boldsymbol{v} \in L_1,\ A(\boldsymbol{x}) = A(\boldsymbol{u}) + A(\boldsymbol{v}) = \boldsymbol{0} + A'(\boldsymbol{v})$. Noticing that $\ker A' = \{\boldsymbol{0}\}, \operatorname{Im} A' = \operatorname{Im} A$,

$$\forall \text{ basis } \boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_r \in \operatorname{Im} A,\ \exists \boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_r \in L_1, \boldsymbol{f_i} = A'(\boldsymbol{e}_i),\ \text{is also a basis} \Longrightarrow A' : L_1 \xrightarrow{\sim} \operatorname{Im} A.$$

We derived $L_1 \cong \operatorname{Im} A$, and $\dim L = \dim(\ker A) + \dim L_1 = \dim(\ker A) + \dim(\operatorname{Im} A)$.

We select bases $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_r \in L_1$, and $\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_{n-r} \in \ker A$. Combining, we obtain $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_r, \boldsymbol{u}_1, \cdots, \boldsymbol{u}_{n-r}$ as another basis of L.

Since A' is an isomorphism, $A'(\boldsymbol{v}_1), A'(\boldsymbol{v}_2), \cdots, A'(\boldsymbol{v}_r) \in \operatorname{Im} A$, and we just expand these linear independent vectors to a basis in M. Thus, for a basis $\mathcal{F} = A'(\boldsymbol{v}_1), A'(\boldsymbol{v}_2), \cdots, A'(\boldsymbol{v}_r), \boldsymbol{f}_{r+1}, \cdots, \boldsymbol{f}_m \in M$, we found a basis $\mathcal{E} = \boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_r, \boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_{n-r} \in L$ such that it satisfied

$$[A]_{\mathcal{F}\leftarrow\mathcal{E}} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}.$$

□

Corollary 3.30. $\forall A = M^{m\times n}, \exists$ *nonsingular* B, C *such that*

$$BAC^{-1} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}$$

3.4 Dual space

The Dual space was invented to decipher a specific type of operator, $A : L \to \mathbb{K}$, where $\mathbb{K}$ is the field of scalars..

Definition 3.31 (Dual Space)**.** *For a vector space* L, *its dual space is defined as*

$$L^* := \hom L, \mathbb{K} = \{f \mid f : L \to \mathbb{K}, f(\boldsymbol{x} + \boldsymbol{y}) = f(\boldsymbol{x}) + f(\boldsymbol{y}), f(k\boldsymbol{x}) = kf(\boldsymbol{x})\}\,.$$

$\hom X, Y$ in Linear Algebra represents the maps that preserves the property of linearity.

Remark 3.32 (Infinite-Dimensional linearity & Dual Space)**.** *In the history of studying linear functions on infinite-dimensional vector spaces, an impressive idea was that for many* continuous linear functionals, *especially in Hilbert spaces such as* L^2, *continuous linear functionals can often be represented in the prototypical form*

$$f_\phi(\boldsymbol{x}) := \int_a^b \phi(t)\boldsymbol{x}(t)dt.$$

For linear space $L = \left\{\boldsymbol{x} : \boldsymbol{x} \in C^1([a, b]), \boldsymbol{x}(a) = \boldsymbol{x}(b) = 0\right\}$, Paul Dirac *introduced Dirac* δ, *which may formally be viewed as the derivative of the generalized cutoff function, initially lacking proper definition of derivative. However, if we use the prototypical function,*

$$f_{\phi'}(\boldsymbol{x}) := \int_a^b \boldsymbol{x}(t)d\phi(t) = \boldsymbol{x}(t)\phi(t)|_a^b - f_\phi(\boldsymbol{x}') = -f_\phi(\boldsymbol{x}')$$

whenever ϕ *is differentiable. We define the Dirac distribution* δ_{t_0} *through*

$$f_{\delta_{t_0}}(\boldsymbol{x}) := -f_H(\boldsymbol{x}'),$$

which formally corresponds to

$$\delta_{t_0} = \frac{d}{dt}H(t-t_0),$$

and thus, in the distributional sense, $\delta_{t_0} = \frac{d}{dt}H(t-t_0)$.

Now, by Dirac δ *function,* $\forall t_0 \in [a,b]$,

$$f_\delta(\boldsymbol{x}) = -f_H(\boldsymbol{x}') = -\int_a^b H(t-t_0)d\boldsymbol{x}(t) = -\int_{t_0}^b d\boldsymbol{x}(t) = \boldsymbol{x}(t_0).$$

The δ *evaluates the value of* $\boldsymbol{x}$ *at one point.*

From the dual-space perspective, the function ϕ *may be viewed as a* weight *selecting information from* $\boldsymbol{x}$*. The Dirac* δ *reminds me the smooth cutoff function in my Exponential Sum Proofs,* $\eta(t)$.

$$\eta(x) = \begin{cases} 0 & \textit{for} \quad x > C \\ 1 & \textit{for} \quad x < -C \\ C^\infty(\mathbb{R}) & \textit{for} \quad x \in [-C,C] \end{cases}.$$

As well as the supporting function ψ, $\exists$ *constants* $A < B$, $\psi(u) \neq 0 \Leftrightarrow u \in [A,B]; \quad \psi \in C^\infty((A,B))$.

The similarities expose in ways they operates, though one is distributional while the other is smooth. The utility of both functional Dirac δ *and* η, ψ *is to serve as* weight *in the integral.*

From the weight *perspective, we consider Dirac* δ *a non-continuous, singular-point-only weight that only allows one point to pass; while* η, ψ *are absolutely smooth for which they return roughly sum(integration) on certain interval to reduce the error from Fourier transform. Hence, both singular distributions such as* δ *and smooth cutoffs such as* η, ψ *may be interpreted as different kinds of weights acting through an integral pairing.*

Definition 3.33 (Dual Basis)**.** *For a basis* $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$*, its dual basis are vectors* $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n \in L^*$*, such that*

$$\boldsymbol{f}_i(\boldsymbol{e}_j) = \delta_{ij}.$$

δ_{ij} *is* Kronecker delta*.*

Theorem 3.34. *If vector space* L *has a finite dimension,* $\dim L = \dim L^*$.

Proof. We are prone to prove that the Dual basis in the previous definition is an actual basis in the dual space.

$\forall \boldsymbol{f} \in L^*, \boldsymbol{x} \in L$, for a basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$,

$$\begin{aligned} \boldsymbol{f}(\boldsymbol{x}) &= a_1\boldsymbol{f}(\boldsymbol{e}_1) + a_2\boldsymbol{f}(\boldsymbol{e}_2) + \cdots + a_2\boldsymbol{f}(\boldsymbol{e}_n) \\ &= a_1b_1 + a_2b_2 + \cdots + a_nb_n \quad (b_i = f(\boldsymbol{x}_i)). \end{aligned}$$

Incorporating Dual basis,

$$\begin{aligned} \boldsymbol{f}_i(\boldsymbol{x}) &= a_1\boldsymbol{f}_i(\boldsymbol{e}_1) + a_2\boldsymbol{f}_i(\boldsymbol{e}_2) + \cdots + a_n\boldsymbol{f}_i(\boldsymbol{e}_n) \\ &= a_i \\ \Longrightarrow \boldsymbol{f}(\boldsymbol{x}) &= b_1\boldsymbol{f}_1(\boldsymbol{x}) + b_2\boldsymbol{f}_2(\boldsymbol{x}) + \cdots + b_n\boldsymbol{f}_n(\boldsymbol{x}) \iff \boldsymbol{f} = b_1\boldsymbol{f}_1 + b_2\boldsymbol{f}_2 + \cdots + b_n\boldsymbol{f}_n. \end{aligned}$$

If $\boldsymbol{f} = 0$, then $\forall \boldsymbol{x} \in L, \boldsymbol{f}(\boldsymbol{x}) = 0$, which only happens when $b_1, b_2, \cdots, b_n$ are all 0. Thus, $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n$ are linear independent. Plus, $\forall \boldsymbol{f} \in L^*$, $\boldsymbol{f}$ can be represented by linear combination of a dual basis. Hence, any dual basis is an actual basis, $\dim L = \dim L^*$. $\square$

Corollary 3.35. *If vector space* L *has a finite dimension,* $L \cong L^*$*. However, if we are considering infinitely dimensional cases,* L *and* L^* *are not necessarily isomorphic.*

We haven't yet noticed the dual relationship between the elements in L and mapping in L^*. To unravel the inner attribute of dual, we are firstly interrupted by the fact that the creation of isomorphic mapping between L and L^* cannot be done without the selection of a basis in L ($\nexists$ canonical isomorphism). Instead of studying directly L, L^*, we try to discover a $A : L \xrightarrow{\sim} L^{**}, L^{**} := (L^*)^*$ that does not rely on basis.

Lemma 3.36. $L \cong L^*$ *as vector spaces.*

Proof. The space L^{**} is the set of linear functionals on L^* that map $f \in L^*$ to $k \in \mathbb{K}$. The $A(\boldsymbol{x})$ is functional; considering

$$A(\boldsymbol{x})[\boldsymbol{f}] := \boldsymbol{f}(\boldsymbol{x}) \quad (\boldsymbol{x} \in L, \boldsymbol{f} \in L^*),$$

it is trivial to prove its linearity.

For basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n \in L$, its dual basis $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n \in L^*$, and the dual basis of the dual basis in L^*, $\boldsymbol{G}_1, \boldsymbol{G}_2, \cdots, \boldsymbol{G}_n \in L^{**}$,

$$\boldsymbol{f_i}(\boldsymbol{e}_j) = \delta_{ij},\ \boldsymbol{G}_i[\boldsymbol{f}_j] = \delta_{ij}.$$

If we can verify that $A(\boldsymbol{e}_i) = \boldsymbol{G}_i$, then the bases of L, L^{**} are one-to-one corresponded, and thus $L \cong L^{**}$, but

$$A(\boldsymbol{e}_i)[\boldsymbol{f}_j] = \boldsymbol{f}_j(\boldsymbol{e}_i) = \delta_{ij} = \boldsymbol{G}_i[\boldsymbol{f}_j] \Longrightarrow A(\boldsymbol{e}_i) = \boldsymbol{G}_i.$$

□

Definition 3.37. *We define* $\langle \boldsymbol{f}, \boldsymbol{x} \rangle := \boldsymbol{f}(\boldsymbol{x})$, $\boldsymbol{x} \in L, \boldsymbol{f} \in L^*$. *It is another notation for* $A(\boldsymbol{x})[\boldsymbol{f}]$.

If we treat $\langle \boldsymbol{f}, \boldsymbol{x} \rangle$ as a "function" for both $\boldsymbol{x}$(as a function) and $\boldsymbol{f}$(as a functional), we find that the $\langle \boldsymbol{f}, \boldsymbol{x} \rangle$ satisfies

1. $\langle \boldsymbol{f}, \boldsymbol{x}_1 + \boldsymbol{x}_2 \rangle = \langle \boldsymbol{f}, \boldsymbol{x}_1 \rangle + \langle \boldsymbol{f}, \boldsymbol{x}_2 \rangle,\ \langle \boldsymbol{f}, k\boldsymbol{x} \rangle = k \langle \boldsymbol{f}, \boldsymbol{x} \rangle,$
2. $\langle \boldsymbol{f}_1 + \boldsymbol{f}_2, \boldsymbol{x} \rangle = \langle \boldsymbol{f}_1, \boldsymbol{x} \rangle + \langle \boldsymbol{f}_2, \boldsymbol{x} \rangle,\ \langle k\boldsymbol{f}, \boldsymbol{x} \rangle = k \langle \boldsymbol{f}, \boldsymbol{x} \rangle,$
3. $\forall \boldsymbol{x} \in L, \langle \boldsymbol{f}, \boldsymbol{x} \rangle = 0, \boldsymbol{f} = \boldsymbol{0}, \forall \boldsymbol{f} \in L^*, \langle \boldsymbol{f}, \boldsymbol{x} \rangle = 0, \boldsymbol{x} = \boldsymbol{0}$ (nondegeneracy).

It is a vivid example of bilinearity.

Definition 3.38 (Annihilator)**.** $L' \subset L$, *the annihilator*

$$(L')^\circ := \{\boldsymbol{f} : \boldsymbol{f} \in L^*, \forall \boldsymbol{x} \in L', \boldsymbol{f}(\boldsymbol{x}) = 0\}.$$

Also denoted by $\operatorname{Ann}(W)$.

Lemma 3.39. *For vector spaces* $L' \subset L, L^*,\ \dim(L')^\circ = \dim L - \dim L'$.

Proof. For the basis $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_r \in L'$, we expand the basis to the L, $\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_r, \boldsymbol{e}_{r+1}, \cdots, \boldsymbol{e}_n$. Its dual basis is $\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_r, \boldsymbol{f}_{r+1}, \cdots, \boldsymbol{f}_n$.

Obviously, the basis of $(L')^\circ$ is $\boldsymbol{f}_r, \boldsymbol{f}_{r+1}, \cdots, \boldsymbol{f}_n$, the *complementary set* of the corresponding dual basis. Otherwise, there is a $\boldsymbol{f}_i \in (L')^\circ, \exists \boldsymbol{x} \in L', \boldsymbol{f}_i(\boldsymbol{x}) \neq 0$. □

Corollary 3.40. *For vector spaces* $L' \subset L, L^*,\ L' \cong ((L')^\circ)^\circ$.

From our understanding of the complementary nature of the annihilator, we reveal a fact:

Lemma 3.41. $\forall L_1, L_2 \subset L,\ (L_1 + L_2)^\circ = L_1^\circ \cap L_2^\circ$.

Theorem 3.42 (Dual Principle (Non-rigorous))**.** *For a finite-dimensional vector space L, if a relevant theorem is delivered only through concepts of* Subspaces, Dimensions, Sums, Intersections*, then another theorem, which is constructed by replacing*

$$\begin{array}{rcl} \textit{Original Theorem } L & \longrightarrow & \textit{Dual Theorem } L^* \\ \textit{Subspace } L_1 & \longrightarrow & L_1^\circ \\ L_1 \subset L_2 & \longrightarrow & L_2^\circ \subset L_1^\circ \\ L_1 + L_2 & \longrightarrow & L_1^\circ \cap L_2^\circ \\ L_1 \cap L_2 & \longrightarrow & L_1^\circ + L_2^\circ \\ 0 & \longrightarrow & L^* \\ L & \longrightarrow & 0 \end{array}$$

is also valid.

If we translate a theorem to its dual by taking the annihilator of all its subjects, we eventually obtain a new corresponding theorem.

For an operator $A \in \hom L, M$, we define its dual operator.

Definition 3.43 (Dual Map)**.** $\forall A \in \hom L, M,\ \exists! A^* \in \hom M^*, L^*$ *such that* $\forall \boldsymbol{g} \in M^*$,

$$(A^*(\boldsymbol{g}))(\boldsymbol{x}) = (\boldsymbol{g} \circ A)(\boldsymbol{x}) \iff \langle A^*(\boldsymbol{g}), \boldsymbol{x} \rangle = \langle \boldsymbol{g} \circ A, \boldsymbol{x} \rangle\,, \quad \boldsymbol{x} \in L$$

Theorem 3.44 (Basis Shifts regarding dual spaces)**.** *For any bases* $\mathcal{E} \in L, \mathcal{F} \in M$ *and their dual bases* $\mathcal{E}' \in L^*, \mathcal{F}' \in M^*$, *an operator* $A \in \hom L, M$ *and its dual map* $A^* \in \hom M^*, L^*$ *satisfy*

$$[A]_{\mathcal{F} \leftarrow \mathcal{E}} = [A^*]^T_{\mathcal{E}' \leftarrow \mathcal{F}'}.$$

Proof. We examine the matrix by using the precedent theorem of basis shift.

$$\begin{cases} A(\boldsymbol{e}_1) = a_{11}\boldsymbol{f}_1 + a_{21}\boldsymbol{f}_2 + \cdots + a_{m1}\boldsymbol{f}_m \\ A(\boldsymbol{e}_2) = a_{12}\boldsymbol{f}_1 + a_{22}\boldsymbol{f}_2 + \cdots + a_{m2}\boldsymbol{f}_m \\ \cdots \\ A(\boldsymbol{e}_n) = a_{1n}\boldsymbol{f}_1 + a_{2n}\boldsymbol{f}_2 + \cdots + a_{mn}\boldsymbol{f}_m \end{cases}, \qquad [A]_{\mathcal{F} \leftarrow \mathcal{E}} := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

in which $\mathcal{E} = (\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n)$, $\mathcal{F} = (\boldsymbol{f}_1, \boldsymbol{f}_2, \cdots, \boldsymbol{f}_n)$.

Then, for the A^*, we have to scrutinize how it transforms $\boldsymbol{f'} \in \mathcal{F}'$ to $\boldsymbol{e'} \in \mathcal{E}'$. $\forall \boldsymbol{e_k} \in \mathcal{E}$,

$$\begin{cases} \langle A^*(\boldsymbol{f}'_1), \boldsymbol{e}_k \rangle = \langle b_{11}\boldsymbol{e}'_1 + b_{21}\boldsymbol{e}'_2 + \cdots + b_{m1}\boldsymbol{e}'_m,\ \boldsymbol{e_k} \rangle = b_{k1} \\ \langle A^*(\boldsymbol{f}'_2), \boldsymbol{e}_k \rangle = \langle b_{12}\boldsymbol{e}'_1 + b_{22}\boldsymbol{e}'_2 + \cdots + b_{m2}\boldsymbol{e}'_m,\ \boldsymbol{e_k} \rangle = b_{k2} \\ \cdots \\ \langle A^*(\boldsymbol{f}'_n), \boldsymbol{e}_k \rangle = \langle b_{1n}\boldsymbol{e}'_1 + b_{2n}\boldsymbol{e}'_2 + \cdots + b_{mn}\boldsymbol{e}'_m,\ \boldsymbol{e_k} \rangle = b_{kn} \end{cases}$$
$$\begin{aligned} \Longrightarrow \langle A^*(\boldsymbol{f}'_i), \boldsymbol{e}_k \rangle &= \langle \boldsymbol{f}'_i \circ A, \boldsymbol{e}_k \rangle = \boldsymbol{f}'_i(a_{1k}\boldsymbol{f}_1 + a_{2k}\boldsymbol{f}_2 + \cdots + a_{mk}\boldsymbol{f}_m) \\ &= a_{ik} = b_{ki}, \end{aligned}$$

which means $(a_{ij}) = (b_{ij})^T$, but they are $[A]_{\mathcal{F} \leftarrow \mathcal{E}}$ and $[A^*]^T_{\mathcal{E}' \leftarrow \mathcal{F}'}$. ∎

4 A Short summary from Gaussian elimination to vector space

Theorem 4.1 (1605, Binet-Cauchy Formula)**.** $A \in M^{m \times n}, B \in M^{n \times m}$.

$$C = AB = (c_{ij}), \quad c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \quad (i, j = 1, \ldots, m),$$

$A_{i_1,\ldots,i_m}$ *selects* i *columns of matrix* A*;* $B_{i_1,\ldots,i_m}$ *extracts* B*'s columns.*

$$\det(C) = \sum_{1 \leq i_1 < \cdots < i_m \leq n} \det A_{i_1,\ldots,i_m} \det B_{i_1,\ldots,i_m}.$$

If $m > n$, $\det C = 0$.

Proof. Denote by

$$\mathcal{I} = \{(i_1, \ldots, i_m) : 1 \leq i_1 < \cdots < i_m \leq n\}.$$

1. **Leibniz expansion of** $\det C$**.**

   Using Leibniz formula,

   $$\det C = \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{l=1}^{m} c_{l\sigma(l)} = \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{l=1}^{m} \sum_{k=1}^{n} a_{lk} b_{k\,\sigma(l)}.$$
   $$\det C = \sum_{\sigma \in S_m} \varepsilon(\sigma) \sum_{1 \leqslant k_1,\ldots,k_m \leqslant n} \prod_{l=1}^{m} a_{lk_l} b_{k_l\,\sigma(l)},$$

   by expanding the product.

2. **Cancellation of terms with repeated indices.**

   If $(k_1, \dots, k_m)$ contains repeated values, say $k_p = k_q$ with $p \neq q$, then exchanging p and q does not change the product
$$\prod_{l=1}^{m} a_{lk_l} b_{k_l\,\sigma(l)},$$
   but changes the sign of the permutation σ. Hence such terms cancel in pairs.

   Therefore, only tuples $(k_1, \dots, k_m)$ with all k_l distinct contribute.

3. **Reindexing by subsets and permutations.**

   If $k_1, \dots, k_m$ are all distinct, then they form a subset $I = \{i_1, \dots, i_m\} \in \mathcal{I}$, and there exists a unique permutation $\tau \in S_m$ such that $k_l = i_{\tau(l)}$. Thus, we may rewrite
$$\sum_{k_1,\dots,k_m \text{ distinct}} \cdot \iff \sum_{I\in\mathcal{I}} \sum_{\tau\in S_m} \cdot$$
$$\det C = \sum_{\sigma\in S_m} \varepsilon(\sigma) \sum_{I\in\mathcal{I}} \sum_{\tau\in S_m} \prod_{l=1}^{m} a_{l\,i_{\tau(l)}} b_{i_{\tau(l)}\,\sigma(l)}.$$

   Let $\rho := \sigma \circ \tau^{-1}$. Then $\sigma = \rho \circ \tau$, and $\varepsilon(\sigma) = \varepsilon(\rho)\varepsilon(\tau)$. Hence,
$$\det C = \sum_{I\in\mathcal{I}} \sum_{\rho\in S_m} \varepsilon(\rho) \left(\sum_{\tau\in S_m} \varepsilon(\tau) \prod_{l=1}^{m} a_{l\,i_{\tau(l)}} \right) \left(\prod_{k=1}^{m} b_{i_k\,\rho(k)} \right).$$

4. **Identification with determinants.**

   Observe that
$$\sum_{\tau\in S_m} \varepsilon(\tau) \prod_{l=1}^{m} a_{l\,i_{\tau(l)}} = \det(A_{i_1,\dots,i_m}),$$
$$\sum_{\rho\in S_m} \varepsilon(\rho) \prod_{k=1}^{m} b_{i_k\,\rho(k)} = \det(B_{i_1,\dots,i_m}).$$

   Therefore,
$$\det C = \sum_{I\in\mathcal{I}} \det(A_{i_1,\dots,i_m}) \det(B_{i_1,\dots,i_m}).$$

If $m > n$, there are no such subsets I, hence $\det C = 0$. $\square$